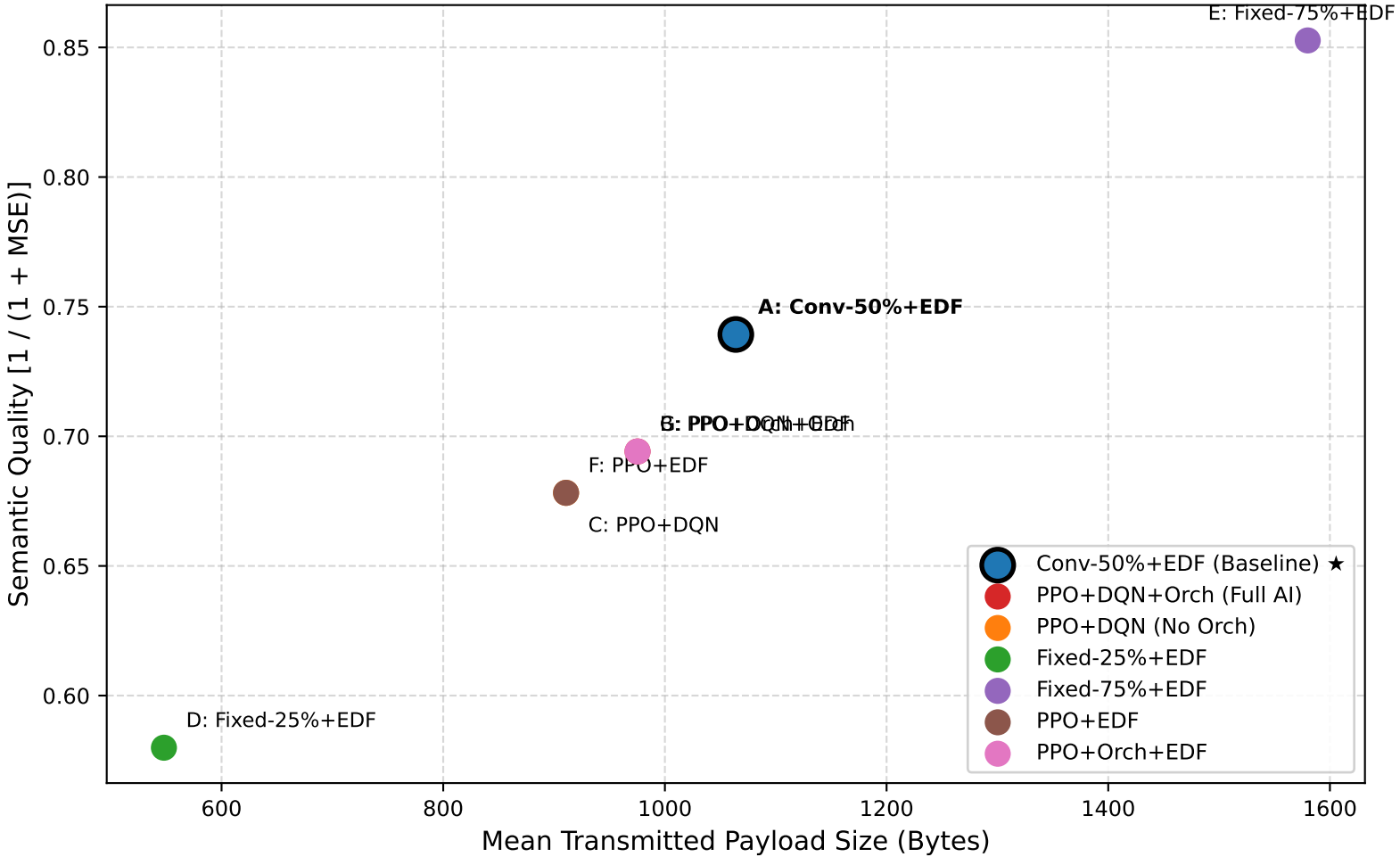
Rate-Distortion Frontier: Semantic Quality vs Payload Size
Semantic Quality [1 / (1 + MSE)]
Mean Transmitted Payload Size (Bytes)
0.85
0.80
0.75
0.70
0.65
0.60
600
800
1000
1200
1400
1600
E: Fixed-75%+EDF
A: Conv-50%+EDF
F: PPO+EDF
C: PPO+DQN
D: Fixed-25%+EDF
Conv-50%+EDF (Baseline) ★
PPO+DQN+Orch (Full AI)
PPO+DQN (No Orch)
Fixed-25%+EDF
Fixed-75%+EDF
PPO+EDF
PPO+Orch+EDF